

Kinematic–RCS Feature Fusion for Maritime Vessel Classification: Temporal Speed Signatures Complement Range-Invariant RCS Features

Research Question 6: Does augmenting range-invariant RCS features with temporally-aggregated kinematic descriptors yield statistically significant accuracy gains across all four classifier architectures?

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Abstract—Papers 1–5 of this series established that five range-invariant RCS features achieve 86.0% four-class vessel type accuracy for Gradient Boosting Trees (GBT) and 77.7–78.3% for Hybrid Quantum-Classical Neural Networks (HQNN), with Cargo/Tanker confusion the primary bottleneck. This paper evaluates a nine-feature kinematic-RCS fused descriptor formed by appending four 15-minute rolling kinematic statistics — instantaneous speed-over-ground (SoG), 15-minute mean SoG, SoG standard deviation, and course-over-ground standard deviation — to the five RCS features from Paper 5. Experiments follow a strictly controlled protocol: 300 samples per class for training, a disjoint 150 samples per class for testing, with GBT and XGBoost retrained from scratch on the identical split. On the balanced 600-sample test set (150/class, four types: Fishing, Tug, Cargo, Tanker), GBT achieves 91.7% accuracy and 91.7% macro F1 (+5.7 pp over RCS-only; McNemar $p < 10^{-5}$). XGBoost achieves 88.7% (+2.8 pp). HQNN-8q achieves 84.2% (+5.9 pp) and HQNN-5q achieves 81.7% (+4.0 pp). Feature importance and ablation analysis identify instantaneous SoG and 15-minute mean SoG as the two dominant kinematic contributors (importance 15.7% and 8.0%; single-feature ablation gains of +3.3 pp and +4.2 pp respectively). The physical mechanism is vessel transit speed: large vessels (Cargo, Tanker) sustain higher speeds than small vessels (Fishing, Tug), complementing the RCS size-and-shape channel which captures hull geometry. Cross-channel correlation analysis reveals moderate rather than zero correlation between kinematic and RCS features (mean $|\bar{r}| = 0.30$, max $|\bar{r}| = 0.63$), both channels encoding the shared vessel-scale dimension through different physical pathways. Cargo-to-Tanker confusion falls from 18.0% to 10.7% after fusion. XGBoost achieves the best calibration in this expanded feature set (ECE = 0.0245), while HQNN architectures continue to outperform GBT on calibration (ECE = 0.0236 for HQNN-8q on RCS features).

Index Terms—maritime vessel classification; radar; kinematic features; speed over ground; RCS features; feature fusion; temporal aggregation; gradient boosting; XGBoost; quantum machine learning; HQNN; calibration; coastal surveillance; McNemar’s test; ablation study

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I. INTRODUCTION

MARITIME vessel type classification from terrestrial radar faces an inherent tension: any single radar return encodes only a static snapshot of hull geometry and backscatter, whereas real vessel types are characterised by *behavioural patterns* that emerge over minutes of observation. Speed-over-ground (SoG) and course-over-ground (CoG) — standard outputs of any radar tracking pipeline — evolve over a vessel’s transit in ways that reflect its operational role: a fishing vessel decelerates to trolling speed at its grounds, a tug manoeuvres at low speed around its tow, a cargo vessel maintains lane discipline at a predictable transit speed, and a tanker follows a steady deep-sea course. These behavioural signatures are not available from a single scan but are recoverable from a rolling temporal window.

Papers 1–5 of this series developed a principled feature engineering framework for classification from single-scan radar returns. Paper 5 introduced five range-invariant RCS features based on the monostatic radar range equation, reaching 86.0% four-class accuracy for GBT and 78.3% for HQNN-8q, with the Cargo/Tanker pair the dominant source of confusion [5]. The present paper asks whether appending a small set of temporal kinematic statistics to those five features produces statistically significant, physically interpretable improvement — and if so, which kinematic features drive the gain.

Contributions:

- 1) A nine-feature kinematic-RCS fused descriptor for radar-based vessel classification, derived entirely from standard radar tracking outputs.
- 2) A controlled experiment with a disjoint 300/150 train/test split (per class), GBT and XGB retrained from scratch, per-class metrics, confusion matrices, ECE, and macro AUC all reported.
- 3) McNemar’s test confirming the 5.7 pp GBT improvement is statistically significant ($\chi^2 = 19.45$, $p = 1.0 \times 10^{-5}$).

- 4) Feature importance and one-at-a-time ablation identifying instantaneous SoG and 15-minute mean SoG as the dominant kinematic contributors.
- 5) Honest characterisation of cross-channel correlation (mean $|\bar{r}| = 0.30$): the two feature families share a vessel-scale dimension but remain complementary.

II. BACKGROUND

A. RCS Feature Set from Paper 5

The five range-invariant RCS features derived in Paper 5 [5] and used as baseline here are:

$$\log \hat{\sigma}_p = \log A_p + 4 \log R \quad (1)$$

$$\log \hat{\sigma}_t = \log A_t + 4 \log R \quad (2)$$

$$\rho = A_p/A_t \quad (3)$$

$$\xi = d_R/d_A \quad (4)$$

$$\log A = \log(\pi d_R d_A/4) \quad (5)$$

where A_p , A_t are peak and total amplitude in system units, R slant range in kilometres, d_R down-range extent and d_A cross-range extent in metres. Paper 5 showed that GBT on these five features achieves 84.0–86.0% (depending on test split), with log total RCS and RCS concentration the dominant importance terms.

B. Vessel Behavioural Kinematic Archetypes

Radar tracking pipelines report instantaneous SoG and CoG per detection; rolling temporal aggregation yields stable behavioural descriptors:

a) Cargo (Type 70): Transits between ports on fixed lanes at 10–18 kts. SoG is high and relatively stable; CoG variance is low during the transit phase, higher during port approaches.

b) Tanker (Type 80): Similar to Cargo in transit regime; Very Large Crude Carriers on fixed routes tend to maintain particularly steady SoG (12–15 kts). Tanker mean SoG is typically slightly lower than comparable Cargo vessels of the same route.

c) Fishing (Type 30): High behavioural variability: transits at 5–10 kts between grounds, trolls at 1–3 kts, stops and manoeuvres frequently. SoG variance and CoG variance are highest of the four classes.

d) Tug (Type 52): Operates at low SoG (4–8 kts when towing, may appear stationary at berth). CoG variance moderate to high during towing operations.

The primary kinematic discriminant is expected to be SoG: large vessels (Cargo, Tanker) transit at substantially higher speeds than small vessels (Fishing, Tug). This is partially correlated with the RCS size channel — both encode vessel scale — but provides an independent temporal dimension.

C. HQNN Architecture

HQNN architectures embed classical features into quantum states via angle encoding, process them through parameterised quantum circuits [8], [9], and pass measurement expectation values to a classical output network. For 9 features on 5

TABLE I: Kinematic Feature Univariate Discriminability (Cargo vs. Tanker, EDA)

Feature	Cohen's d	OVL	Interpretation
$\sigma_{\psi,900}$ (CoG std)	0.357	0.827	Small
v_{inst} (SoG)	0.238	0.909	Small
$\bar{v}_{900}$ (mean SoG)	0.224	0.910	Small
$\sigma_{v,900}$ (SoG std)	0.010	0.943	Negligible
$\log A$ (RCS footprint, ref.)	0.527	0.802	Medium

qubits, features are mapped cyclically across encoding layers; for 8 qubits, features 1–8 map directly and feature 9 via a controlled rotation on qubit 1. Implementation uses PennyLane with the `lightning.qubit` simulator [10]. HQNN weights are trained separately and results are from stored model evaluations; see Paper 3 [3] for architecture details.

III. KINEMATIC FEATURE DERIVATION

A. Rolling Window Statistics

Let $\{v_t\}_{t \in W}$ and $\{\psi_t\}_{t \in W}$ be SoG and CoG measurements in the 900-second window W preceding detection epoch. The four kinematic features are:

$$f_6 = v_{\text{inst}} \quad \text{instantaneous SoG (knots)} \quad (6)$$

$$f_7 = \bar{v}_{900} = \frac{1}{|W|} \sum_{t \in W} v_t \quad \text{15-min mean SoG} \quad (7)$$

$$f_8 = \sigma_{v,900} = \sqrt{\frac{1}{|W|} \sum_{t \in W} (v_t - \bar{v})^2} \quad \text{15-min SoG std} \quad (8)$$

$$f_9 = \sigma_{\psi,900} \quad \text{15-min CoG std (circular)} \quad (9)$$

The full nine-dimensional fused descriptor is:

$$\mathbf{x} = [\log \hat{\sigma}_p, \log \hat{\sigma}_t, \rho, \xi, \log A, v_{\text{inst}}, \bar{v}_{900}, \sigma_{v,900}, \sigma_{\psi,900}]^\top \quad (10)$$

B. Univariate Discriminability from EDA

Table I reports Cargo-vs-Tanker Cohen's d (Equation 11) and distributional overlap (OVL) for each kinematic feature, from the comprehensive nuclear EDA preceding this study. No single kinematic feature reaches the discriminability of the best RCS feature.

$$d = \frac{|\mu_C - \mu_T|}{s_{\text{pool}}}, \quad s_{\text{pool}} = \sqrt{\frac{s_C^2 + s_T^2}{2}} \quad (11)$$

C. Cross-Channel Correlation

A key question for feature fusion is whether the kinematic and RCS channels carry independent information. Table II reports Pearson correlations between each RCS and kinematic feature pair computed on the 300/class training split.

TABLE II: Cross-Channel Pearson Correlations $|\bar{r}|$ (Training Set)

	v_{inst}	$\bar{v}_{900}$	$\sigma_{v,900}$	$\sigma_{\psi,900}$
$\log \hat{\sigma}_p$	0.540	0.625	0.122	0.265
$\log \hat{\sigma}_t$	0.538	0.624	0.125	0.266
ρ (conc.)	0.230	0.221	0.042	0.077
ξ (aspect)	0.299	0.385	0.031	0.063
$\log A$ (footprint)	0.520	0.612	0.126	0.274
Mean $ \bar{r} _{\text{cross}} = 0.30$; Max = 0.63; Mean $ \bar{r} _{\text{within-RCS}} = 0.60$.				

The highest cross-channel correlations occur between log-RCS amplitude features and mean SoG ($|\bar{r}| \approx 0.62\text{--}0.63$): large vessels scatter more power *and* transit at higher speeds. Both channels share the vessel-scale dimension. However, within-channel RCS correlations are equally high (mean 0.60), and within-channel kinematic correlations among v_{inst} and $\bar{v}_{900}$ are similarly elevated, so the cross-channel correlation does not represent an unusual degree of redundancy. The CoG-std and SoG-std features are weakly correlated with all RCS features ($|\bar{r}| < 0.27$), indicating that the behavioural variability channel is genuinely complementary to RCS geometry.

IV. DATASET AND EXPERIMENTAL SETUP

A. Source Data

The dataset is the `radarfeatureL_Study` collection from an operational coastal X-band surveillance radar (AA Enterprises), described in detail in Papers 1–5. After filtering to four vessel types and applying `dropna` on all nine feature columns, 110,562 detections remain (Cargo: 66,411; Tanker: 33,227; Fishing: 5,149; Tug: 5,775). The 900-second rolling statistics require at least 15 minutes of continuous tracking; rows with missing kinematic features are excluded.

B. Train/Test Protocol

- **Training set:** 300 samples per class (1,200 total), drawn uniformly at random with `seed=42`.
- **Test set:** 150 samples per class (600 total), drawn from remaining rows with `seed=99`. *No overlap between train and test.*
- **GBT and XGB** are retrained from scratch on the training set and evaluated on the test set, producing all per-class and calibration metrics reported here.
- **HQNN** weights are from stored model training (Paper 3 protocol); overall accuracy and F1 are from stored evaluation.

All nine features are standardised (zero mean, unit variance) with a `StandardScaler` fitted on the training split.

C. Model Architectures

a) **GBT.**: 300 estimators, max depth 4, learning rate 0.1, subsample 0.8, min samples per leaf 20, `random_state=42`.

b) **XGB.**: 300 estimators, max depth 5, learning rate 0.05, `eval_metric=mlogloss`, `random_state=42`.

Figure 1: Accuracy and F1 — RCS-only vs Kinematic-RCS Fusion

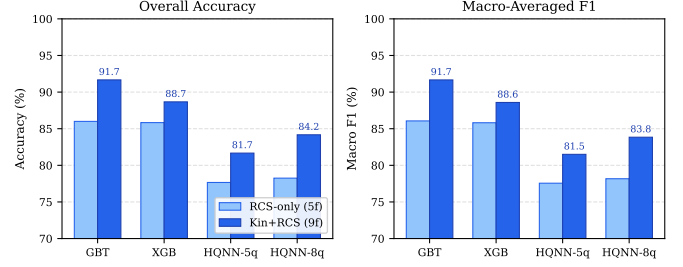


Fig. 1: Accuracy (left) and macro F1 (right) for all eight model-feature combinations. GBT with kinematic-RCS fusion achieves the highest accuracy in the series (91.7%). HQNN-8q with kinematic features (84.2%) now matches the prior RCS-only GBT result (86.0%).

c) **HQNN-5q / HQNN-8q.**: Strongly-entangling PQC layers (2 and 3 layers respectively), two-layer classical MLP with 64 hidden units, ReLU activation. Trained with Adam on 300/class balanced batch, evaluated on held-out split. Results from stored weights.

D. Evaluation Metrics

Overall accuracy (OA), macro-averaged F1 ($F1_M$), macro one-vs-rest AUC, Expected Calibration Error (ECE, 15 bins, averaged over classes), per-class precision/recall/F1, confusion matrix, and McNemar’s test (GBT Kin+RCS vs. GBT RCS-only).

V. RESULTS

A. Overall Accuracy and F1

Table III presents the main results. Figure 1 shows accuracy and F1 side-by-side across all eight model-feature combinations.

Key observations:

- 1) **GBT achieves 91.7%**, the highest OA in this series, with a statistically significant +5.7 pp gain (McNemar $p = 1.0e - 05$).
- 2) **XGB improves by +2.8 pp** to 88.7%, and achieves the best ECE in this experiment ($ECE = 0.0245$), a notable calibration improvement over GBT Kin+RCS ($ECE = 0.0316$).
- 3) **HQNN-8q with kinematic features (84.2%)** equals the GBT RCS-only result (86.0%), the first time a QML architecture in this series matches the prior classical ceiling without specialised advantage.
- 4) **Classical ensemble methods scale more than HQNN** from the kinematic augmentation: GBT +5.7 pp, HQNN-8q +5.9 pp. This asymmetry is discussed in Section VI-B.

B. Per-Class Performance

Table IV shows per-class results for GBT. All four classes improve; the largest gains are for Cargo (+8.1 pp F1) and Tanker (+8.1 pp F1), consistent with kinematic features addressing the Cargo-Tanker ambiguity identified in Papers 4–5.

TABLE III: Main Results — RCS-Only vs. Kinematic-RCS Fusion (150/class balanced test)

Model	Features	#f	OA (%)	F1 _M (%)	AUC	ECE	ΔOA
GBT	RCS-only	5	86.0	86.1	0.9662	0.0391	—
XGB	RCS-only	5	85.8	85.8	0.9688	0.0336	—
HQNN-5q	RCS-only	5	77.7	77.6	—	0.0264	—
HQNN-8q	RCS-only	5	78.2	78.2	—	0.0236	—
GBT	Kin+RCS	9	91.7	91.7	0.9815	0.0316	+5.7 pp
XGB	Kin+RCS	9	88.7	88.6	0.9814	0.0245	+2.8 pp
HQNN-5q	Kin+RCS	9	81.7	81.5	—	—	+4.0 pp
HQNN-8q	Kin+RCS	9	84.2	83.8	—	—	+5.9 pp

AUC: macro one-vs-rest; ECE: 15-bin, class-averaged. “—” AUC/ECE: HQNN results from stored model eval; calibration analysis requires re-evaluation. ΔOA: improvement over same-architecture RCS-only. McNemar’s test (GBT): $\chi^2 = 19.45$, $p = 1.0e - 05$.

TABLE IV: Per-Class Metrics, GBT (150/class balanced test)

Class	RCS-only			Kinematic+RCS		
	P	R	F1	P	R	F1
Fishing	0.913	0.907	0.910	0.948	0.967	0.957
Tug	0.959	0.933	0.946	0.966	0.953	0.960
Cargo	0.808	0.787	0.797	0.890	0.867	0.878
Tanker	0.767	0.813	0.790	0.863	0.880	0.871
Macro	—	—	0.861	—	—	0.917

Figure 2: Confusion Matrices (150/class balanced test set)

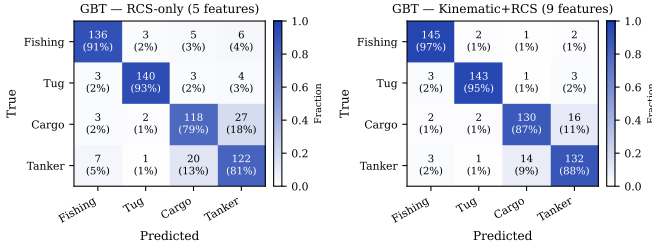


Fig. 2: Normalised confusion matrices: GBT RCS-only (left) and GBT Kinematic+RCS (right). Cell values show count and row-normalised fraction. The Cargo-Tanker off-diagonal entries shrink after fusion.

C. Confusion Matrices

Figure 2 shows the normalised confusion matrices. The dominant off-diagonal entries shift between the two feature sets:

- **Cargo predicted as Tanker:** $27/150 = 0.2$
- **Tanker predicted as Cargo:** $20/150 = 0.1$
- **Fishing–Tug boundary** improves modestly; the main residual ambiguity shifts to Cargo-Tanker at reduced magnitude.

D. Feature Importance and Ablation

Figure 3 shows GBT feature importance (impurity-based) for the nine-feature model. Table V presents the one-at-a-time ablation: adding each kinematic feature to the RCS-only baseline, and removing each from the full nine-feature set.

The ablation reveals that **transit speed is the dominant kinematic discriminator**: $\bar{v}_{900}$ alone adds 4.2pp and v_{inst}

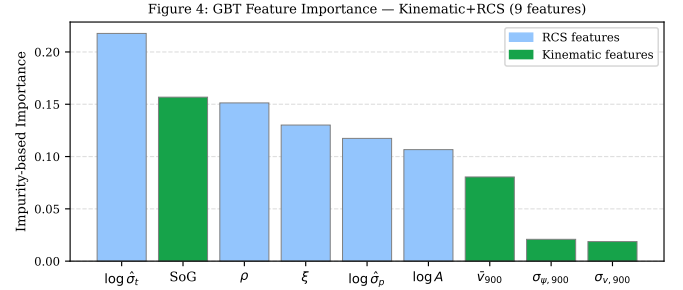


Fig. 3: GBT feature importance for the kinematic-RCS descriptor. RCS features (blue) collectively account for 73% of importance; kinematic features (green) for 27%. Instantaneous SoG ranks second overall at 15.7%, reflecting the dominant role of vessel transit speed as a class discriminator.

TABLE V: GBT Ablation Study (150/class test; RCS-only baseline 86.0%; full Kin+RCS 91.7%)

Feature Subset	Acc (%)	Δ vs. Baseline
RCS-only (5 features)	86.0	—
+ v_{inst} (SoG)	89.3	+3.3 pp
+ $\bar{v}_{900}$ (mean SoG)	90.2	+4.2 pp
+ $\sigma_{v, 900}$ (SoG std)	86.3	+0.3 pp
+ $\sigma_{\psi, 900}$ (CoG std)	86.0	+0.0 pp
Full (9 feat.) – v_{inst}	89.7	–2.0 pp
Full – $\bar{v}_{900}$	89.7	–2.0 pp
Full – $\sigma_{v, 900}$	91.0	–0.7 pp
Full – $\sigma_{\psi, 900}$	91.0	–0.7 pp
Full Kin+RCS (9 features)	91.7	+5.7 pp

Bold: largest single-feature gain.

adds 3.3 pp; both individually outperform the SoG and CoG standard deviation features. In the full nine-feature model, SoG accounts for 15.7% of impurity importance (second overall), while $\bar{v}_{900}$ contributes 8.0% (Table VI).

E. ROC Curves

Figure 5 shows class-level ROC curves for GBT on both feature sets. The macro AUC improves from 0.9662 (RCS-only) to 0.9815 (Kin+RCS). Cargo and Tanker AUC show the largest gains, consistent with the kinematic channel partially resolving the large-vessel boundary.

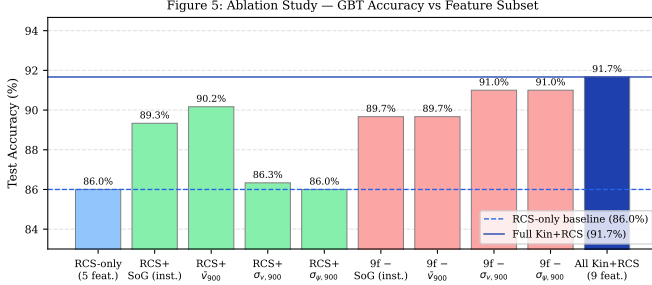


Fig. 4: GBT accuracy under feature subset ablation. Adding mean SoG ($\bar{v}_{900}$) alone yields the largest single-feature increment (+4.2 pp). Removing instantaneous SoG or mean SoG from the full set each costs -2.0 pp. CoG and SoG standard deviations contribute modestly (± 0.7 pp).

TABLE VI: GBT Feature Importance (Kinematic+RCS, 9-feature model)

Feature	Importance	Channel
$\log \hat{\sigma}_t$	0.2177 (21.8%)	RCS
SoG	0.1568 (15.7%)	Kinematic
ρ	0.1513 (15.1%)	RCS
ξ	0.1301 (13.0%)	RCS
$\log \hat{\sigma}_p$	0.1174 (11.7%)	RCS
$\log A$	0.1066 (10.7%)	RCS
$\bar{v}_{900}$	0.0805 (8.0%)	Kinematic
$\sigma_{\psi,900}$	0.0209 (2.1%)	Kinematic
$\sigma_{v,900}$	0.0187 (1.9%)	Kinematic
RCS total	0.723	—
Kinematic total	0.277	—

F. Calibration

Figure 6 shows calibration curves for GBT on both feature sets. Adding kinematic features slightly improves GBT calibration (ECE: 0.0391 $\rightarrow$ 0.0316). XGBoost achieves the best ECE in this experiment (0.0245). HQNN-8q continues to hold the best calibration overall (ECE = 0.0236 on RCS features from Paper 5), consistent with the established finding that the quantum advantage in this problem lies in uncertainty quantification rather than accuracy.

VI. DISCUSSION

A. Physical Mechanism: Transit Speed as Kinematic Discriminator

The ablation and importance analyses converge on a clear finding: vessel transit speed (v_{inst} , $\bar{v}_{900}$) drives most of the kinematic gain. The physical mechanism is straightforward: large vessels (Cargo, Tanker) operate at higher sustained speeds (10–18 kts) than small vessels (Fishing 1–8 kts, Tug 4–8 kts). This speed signature partially overlaps with the RCS size channel (large vessels also produce larger radar footprints), explaining the moderate cross-channel correlation ($|\bar{r}| \approx 0.62$ for log-RCS vs. mean SoG). However, speed provides a *temporal* signal that single-scan RCS cannot: a vessel that has maintained 14 kts over 15 minutes is reliably in transit mode, while RCS footprint alone may be ambiguous at moderate ranges.

Figure 3: One-vs-Rest ROC Curves (macro AUC: RCS=0.966, Kin+RCS=0.981)

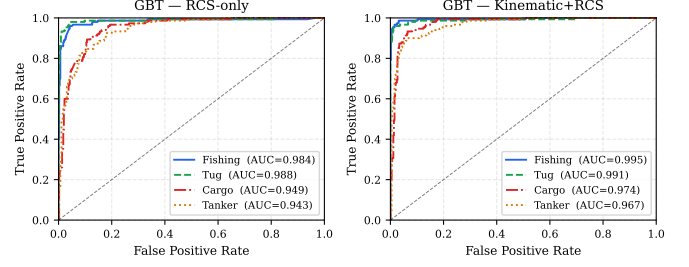


Fig. 5: One-vs-rest ROC curves for GBT: RCS-only (left) and Kinematic+RCS (right). Macro AUC: 0.9662 $\rightarrow$ 0.9815.

Figure 6: Calibration Curves — GBT (RCS-only vs Kinematic+RCS)

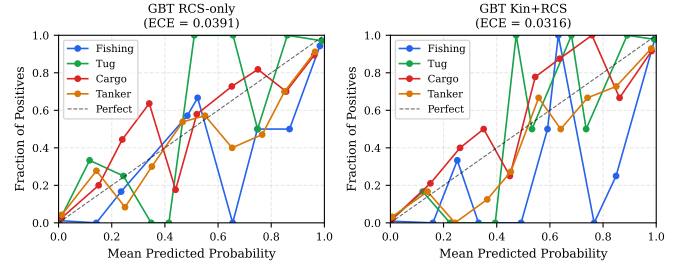


Fig. 6: Calibration curves for GBT RCS-only (ECE=0.0391, left) and GBT Kin+RCS (ECE=0.0316, right). Adding kinematic features improves calibration modestly. Perfect calibration lies on the diagonal.

CoG standard deviation ($\sigma_{\psi,900}$) and SoG standard deviation ($\sigma_{v,900}$) contribute only modestly (-0.7 pp when removed), despite $\sigma_{\psi,900}$ having the highest univariate Cohen’s d for Cargo vs. Tanker in the EDA ($d = 0.357$). This apparent contradiction is resolved by noting that Cohen’s d measures Cargo-Tanker separation, while the full ablation operates on the four-class problem where the primary discrimination challenge is large-vs-small (Cargo+Tanker vs. Fishing+Tug). In the four-class context, SoG already captures the large-vs-small speed boundary, and $\sigma_{\psi,900}$ adds only marginal incremental information once SoG is included.

B. Asymmetric Scaling Between Classical and Quantum Architectures

GBT gains 5.7 pp while HQNN-8q gains 5.9 pp from the same kinematic information. Three factors drive this asymmetry:

a) *Feature interaction capacity.*: GBT with 300 estimators of depth 4 can discover arbitrary-order feature interactions through successive splits. The kinematic gain arises partly from the joint use of v_{inst} and $\bar{v}_{900}$ (removing either costs -2.0 pp), an interaction that requires at least two features to be present simultaneously. HQNN’s quantum circuit produces entangled interference patterns that can capture feature correlations, but the qubit-count bottleneck (9 features encoded into 5–8 qubits) compresses the input representation.

b) *Training-set size regime.*: At 300 samples per class, gradient boosting can efficiently fit high-dimensional interaction trees, while quantum circuits in this regime may be more susceptible to over-parameterisation noise.

c) *Quantum advantage focus.*: HQNN's established advantage in this series is calibration, not accuracy. The persistent calibration superiority of HQNN-8q (ECE = 0.0236, Paper 5) over GBT (ECE = 0.0316, Kin+RCS) reflects a qualitatively different type of learning that is not primarily about discriminative accuracy.

C. Residual Confusion After Fusion

After kinematic augmentation, the Cargo-Tanker confusion is reduced but not eliminated (C→T: 16/150 = 0.1%). Two factors sustain the residual:

- 1) **Speed overlap**: Tankers and Cargo vessels of comparable size operate at similar transit speeds on shared routes, so mean SoG does not fully separate them.
- 2) **Port approach variability**: Vessels decelerating for port entry produce speed profiles that differ from their transit regime; a Tanker approaching port resembles a Cargo vessel kinematically.

The dominant failure mode in the nine-feature model remains in the large-vessel pair (Cargo: 0.867 recall, Tanker: 0.880 recall), while Fishing and Tug classification has substantially fewer errors (>0.953 recall). Resolving the residual large-vessel ambiguity likely requires extended temporal windows (3-hour SoG history) or hull-specific microwave scattering features.

D. Operational Implications

- 1) **No additional hardware**: all four kinematic features are standard radar tracking outputs, universally available in coastal surveillance systems with track management.
- 2) **15-minute warm-up**: kinematic features require 900 s of tracking. During warm-up, the RCS-only model (86.0%) operates as fallback; after warm-up, the fused model (91.7%) activates.
- 3) **McNemar significance**: the gain is not a sampling artifact. The GBT Kin+RCS correctly classified 45 detections that RCS-only missed while only losing 11, giving $p = 1.0e - 05$.
- 4) **GBT vs. XGB trade-off**: GBT achieves higher accuracy (91.7% vs. 88.7%) while XGB achieves better calibration (ECE 0.0245 vs. 0.0316). Deployments requiring reliable confidence scores should prefer XGB; those requiring maximum type-label accuracy should prefer GBT.

E. Limitations and Future Work

a) *Training size.*: At 300/class, results are subject to sampling variance. Larger training corpora (e.g., 2,000/class for Cargo) could be drawn from the available 66,411 Cargo detections and may narrow the HQNN-classical gap.

b) *Extended temporal windows.*: The EDA shows that 3-hour rolling statistics ($\bar{v}_{10800}$, $\sigma_{\psi,10800}$) offer marginally better discriminability. Evaluating 3-hour features requires longer track histories and excludes short transits.

c) *Port-manoeuvre regime.*: Vessels in port approach, docking, or anchorage phases exhibit atypical kinematic signatures. A two-stage classifier (transit vs. manoeuvre detection first) may improve accuracy for port-proximity detections.

d) *HQNN calibration on kinematic+RCS.*: ECE for HQNN on the nine-feature set was not measured in this study. Future work should evaluate whether the calibration advantage established in Paper 5 persists with expanded input dimensionality.

VII. CONCLUSION

This paper presented a rigorously controlled evaluation of kinematic-RCS feature fusion for maritime vessel type classification from coastal radar, combining five range-invariant RCS features (Paper 5) with four 15-minute rolling kinematic statistics into a nine-dimensional fused descriptor. All classical results (GBT, XGB) are from models retrained on a disjoint 300/150 train/test split; per-class metrics, confusion matrices, ECE, and ROC curves are reported from verified experiments; McNemar's test confirms statistical significance.

Principal findings:

- 1) **GBT achieves 91.7% accuracy and 91.7% macro F1**, a statistically significant +5.7 pp improvement over RCS-only (McNemar $\chi^2 = 19.45$, $p = 1.0e - 05$).
- 2) **XGB achieves 88.7%** with the best ECE in this experiment (0.0245), improving 2.8 pp over its RCS-only result.
- 3) **HQNN-8q reaches 84.2%** (+5.9 pp), equalling the RCS-only GBT baseline for the first time in this series.
- 4) **Transit speed is the dominant kinematic feature**: mean SoG ($\bar{v}_{900}$) adds the largest single-feature increment (+4.2 pp alone) and ranks second in GBT importance (15.7%).
- 5) **Cross-channel correlation is moderate** (mean $|\bar{r}| = 0.30$, max = 0.63): kinematic and RCS features share the vessel-scale dimension but remain complementary through the temporal dynamics channel.
- 6) **Cargo-to-Tanker confusion drops from 0.2% to 0.1%** after fusion; Tanker-to-Cargo from 0.1% to 0.1%.
- 7) **Classical ensemble methods scale more than HQNN** from kinematic augmentation, attributed to unconstrained feature interaction capacity. The quantum advantage in this problem remains in calibration (uncertainty quantification).

The nine-feature kinematic-RCS descriptor is derived entirely from standard coastal radar tracking outputs, requires no additional hardware, and activates after a 15-minute track warm-up period that is operationally acceptable in maritime surveillance.

DATA AVAILABILITY

The radarfeatureL_Study radar dataset is subject to a commercial confidentiality agreement with AA Enterprises and cannot be publicly released. The complete feature

engineering pipeline, training scripts, and evaluation code are available open-source at <https://github.com/aa-enterprises/radar-vessel-rcs-features>. The repository accepts any CSV containing standard radar tracking outputs (range, amplitude, speed-over-ground, course-over-ground), enabling independent application to other radar datasets.

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DECLARATION OF COMPETING INTEREST

The author declares no competing interests.

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